



Solution Set

Question 1

Expected exposure is the expected positive mark-to-market of a derivative at a future time. Given $EE(0)=0.0$, $EE(1)=3.2$, $EE(2)=4.1$, $EE(3)=2.0$ million, approximate exposure-years using the trapezoidal rule.

Answer key

With annual spacing $h=1$, the trapezoidal rule gives

$$\begin{aligned}\int_0^3 EE(t) dt &\approx \frac{1}{2}(0 + 3.2) + \frac{1}{2}(3.2 + 4.1) + \frac{1}{2}(4.1 + 2.0) \\ &= 1.6 + 3.65 + 3.05 = 8.30\end{aligned}$$

The result is 8.30 million exposure-years. The units matter: exposure in millions multiplied by time in years.

Question 2

Approximate the discounted exposure integral

$$\int_0^3 e^{-0.04t} EE(t) dt$$

using the same expected-exposure values from Question 1 and the trapezoidal rule.

Answer key

First discount each exposure value:

$$g(t) = e^{-0.04t} EE(t)$$
$$g(0) = 0, \quad g(1) = 3.2e^{-0.04} = 3.0745, \quad g(2) = 4.1e^{-0.08} = 3.7848, \quad g(3) = 2.0e^{-0.12} = 1.7738$$

Now apply trapezoids:

$$\int_0^3 g(t) dt \approx \frac{1}{2}(0 + 3.0745) + \frac{1}{2}(3.0745 + 3.7848) + \frac{1}{2}(3.7848 + 1.7738) = 7.7467$$

Question 3

Use Simpson's 1/3 rule to approximate

$$\int_0^2 e^{-0.05t} dt$$

and compare with the exact value.

Answer key

For two intervals, $h=1$ and

$$S = \frac{h}{3} \{f(0) + 4f(1) + f(2)\}$$
$$S = \frac{1}{3} \{1 + 4e^{-0.05} + e^{-0.10}\} \approx 1.903252$$

The exact value is

$$\int_0^2 e^{-0.05t} dt = \frac{1 - e^{-0.10}}{0.05} \approx 1.903252$$

The agreement is very close because the exponential is smooth over this short interval.

Question 4

Why does the Gaussian normal CDF require numerical integration or approximation? State the integral.

Answer key

The standard normal CDF is

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

The integrand does not have an elementary antiderivative. Risk systems therefore use quadrature, rational approximations, or library routines to compute normal probabilities and VaR quantiles.

Question 5

Derive the change of variables used to compute

$$\mathbb{E}[(S - K)^+]$$

when

$$S$$

is lognormal.

Answer key

The expectation is

$$\mathbb{E}[(S - K)^+] = \int_K^{\infty} (s - K) f_S(s) ds$$

If

$$\log S \sim N(m, v)$$

set

$$z = \frac{\log s - m}{\sqrt{v}}, \quad s = e^{m + \sqrt{v}z}, \quad ds = \sqrt{v} e^{m + \sqrt{v}z} dz$$

This transforms a semi-infinite integral over prices into an integral over normal z -values. Even when a closed form exists for vanilla options, the same technique is crucial for nonstandard payoffs.

Question 6

A desk estimates

$$\mathbb{E}[L \mid L > q]$$

by integrating the tail of a loss density. Explain why grid placement matters.

Answer key

Tail quantities depend on a small, high-impact region of the distribution. A uniform grid may waste many points near the center and undersample the tail. A better grid puts more nodes near q and beyond, or uses a transformation that stretches the tail region.

Question 7

Use the midpoint rule with four intervals to approximate

$$\int_0^1 x^2 dx$$

Include the error direction.

Answer key

The interval width is $h=0.25$ and midpoints are

$$0.125, 0.375, 0.625, 0.875$$

The midpoint estimate is

$$0.25(0.125^2 + 0.375^2 + 0.625^2 + 0.875^2) = 0.328125$$

The exact integral is $1/3 = 0.333333$. Since x^2 is convex, the midpoint rule underestimates the area.

Question 8

Explain the difference between Monte Carlo integration and deterministic quadrature in pricing.

Answer key

Deterministic quadrature uses fixed nodes and weights and can be highly accurate in low dimensions. Monte Carlo integration samples random scenarios and converges more slowly, but it handles high-dimensional path-dependent pricing problems better. This is why vanilla one-dimensional integrals use quadrature, while portfolio simulation often uses Monte Carlo.

Question 9

Show that Simpson's rule is exact for any quadratic polynomial on two equal subintervals.

Answer key

It is enough to check the basis 1 , x , and x^2 on $[-h, h]$. Simpson's rule gives

$$\frac{h}{3} \{f(-h) + 4f(0) + f(h)\}$$

For $f=1$ it gives $2h$, for $f=x$ it gives 0 , and for $f=x^2$ it gives

$$\frac{h}{3}(h^2 + 0 + h^2) = \frac{2h^3}{3}$$

These match the exact integrals over $[-h, h]$. Therefore the rule is exact for any quadratic by linearity.

Question 10

A CVA approximation uses $\sum_i DF_i EE_i PD_i$. Explain this as a numerical integration formula.

Answer key

Continuous-time CVA is an integral of discounted expected exposure against default probability increments:

$$CVA \approx (1 - R) \int_0^T DF(t) EE(t) dPD(t)$$

On a grid, $dPD(t)$ is approximated by default-probability increments PD_i , giving

$$CVA \approx (1 - R) \sum_i DF_i EE_i \Delta PD_i$$

This is numerical integration with respect to a probability measure rather than ordinary calendar time.

Question 11

What convergence check would you run before trusting a numerical integral in a risk report?

Answer key

Refine the grid and verify that the result stabilizes. For example, compare annual, semiannual, quarterly, and monthly grids. If the answer materially changes, the original grid was too coarse or the integrand has features that need special treatment.

Question 12

Why can interpolation dominate integration error in exposure calculations?

Answer key

The exposure is often known only at simulation dates. The integral is then taken over an interpolated curve, not the true curve. If exposure has sharp peaks between grid dates, a highly accurate quadrature rule on a poor interpolation can still give a poor answer.